

BIRSA MUNDA, India

WMO: 427010

Lat: 23.314N

Lon: 85.322E

Elev: 2148

StdP: 13.59

Time Zone: 5.50 (IND)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	47.2	49.4	28.8	24.6	74.4	32.1	28.7	71.3	12.1	70.6	11.1	70.4	1.0	320	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	21.5	103.2	68.6	100.8	68.7	98.4	68.7	79.1	86.9	78.3	85.6	77.6	84.4	7.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	153.7	82.0	76.7	150.8	81.4	76.0	147.1	80.5	44.8	87.2	43.9	86.6	43.2	84.7	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	11.1	9.6	DB	43.3	106.0	1.8	2.4	41.9	107.7	40.9	109.1	39.8	110.4	38.5	112.1	
			WB	40.1	82.2	1.8	2.0	38.8	83.6	37.7	84.8	36.7	85.9	35.3	87.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.7	62.4	68.4	76.5	84.8	87.0	84.5	79.5	79.3	78.9	75.5	69.0	63.1
	DBStd	8.82	3.67	4.68	4.75	4.50	4.32	5.13	2.54	2.19	2.47	3.21	3.35	3.12
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	197	99	18	0	0	0	0	0	0	0	0	5	75
	CDD50	9400	383	516	821	1043	1147	1034	915	908	867	791	569	406
	CDD65	4122	17	114	356	593	682	584	450	443	417	326	124	16
	CDH74	41845	196	1013	3692	7863	9210	6796	3524	3347	3082	2207	768	147
	CDH80	18562	16	257	1705	4585	5421	3442	862	766	835	583	85	5
Wind	WSAvg	3.7	2.9	3.5	4.0	4.5	5.1	4.8	4.6	4.0	3.7	2.5	2.3	2.3
Precipitation	PrecAvg	53.9	0.7	0.7	0.8	0.7	2.6	9.5	13.1	12.4	9.7	3.3	0.4	0.3
	PrecMax	74.2	3.9	2.5	3.1	3.5	5.4	21.3	24.6	22.1	17.3	13.3	2.3	2.2
	PrecMin	34.3	0.0	0.0	0.0	0.0	0.5	2.6	7.0	6.3	1.4	0.0	0.0	0.0
	PrecStd	9.8	0.9	0.7	0.8	0.9	1.5	4.2	4.3	4.2	4.1	3.4	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.2	89.6	98.3	105.5	105.4	105.0	91.6	89.8	90.2	89.7	84.6	80.4
		MCWB	59.1	62.7	63.9	66.4	68.1	69.8	77.2	77.8	76.8	71.4	66.4	61.0
	2%	DB	78.7	86.1	94.4	102.3	103.4	102.2	88.9	87.7	88.1	87.3	82.4	77.4
		MCWB	58.9	61.8	64.2	66.4	68.3	72.1	77.4	77.4	76.2	71.2	65.4	60.0
	5%	DB	75.9	82.9	91.3	100.0	101.3	98.6	86.6	86.0	86.4	85.2	80.6	75.4
		MCWB	57.7	60.8	63.3	66.4	68.8	73.3	77.2	77.1	76.0	71.2	64.3	58.7
	10%	DB	73.4	80.3	88.7	97.2	98.7	94.8	84.6	84.3	84.8	83.3	78.8	73.3
		MCWB	56.3	59.6	62.9	66.2	69.1	74.7	76.8	76.7	75.8	70.8	63.8	57.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	68.3	70.6	76.5	79.3	80.4	79.6	79.7	79.4	77.3	71.7	65.6
		MCDB	70.8	78.7	83.7	89.3	90.0	89.9	86.4	86.0	85.9	83.9	77.7	73.8
	2%	WB	62.6	65.9	68.3	73.9	77.4	78.9	78.6	78.5	78.2	76.2	70.0	63.4
		MCDB	71.4	76.7	83.2	87.5	88.8	88.7	85.2	84.4	84.6	82.4	75.6	71.8
	5%	WB	60.8	63.9	67.0	71.7	76.2	78.0	77.8	77.7	77.3	75.2	68.3	61.6
		MCDB	70.1	75.4	82.5	86.8	88.2	87.3	83.7	83.1	83.3	81.2	75.4	70.8
	10%	WB	59.1	62.3	65.6	70.2	75.0	77.2	77.2	77.1	76.6	74.0	66.7	60.0
		MCDB	68.8	74.2	81.7	86.6	87.3	86.0	82.5	82.2	82.1	79.2	74.8	69.6
Mean Daily Temperature Range	5% DB	MDBR	23.1	22.8	23.2	23.6	21.5	15.6	9.8	9.7	11.1	15.9	20.9	22.4
		MCDBR	25.9	25.3	25.7	25.8	24.2	21.3	13.1	12.4	14.1	17.8	21.5	24.1
		MCWBR	10.5	8.8	7.8	7.9	7.8	6.7	4.4	4.5	5.1	6.4	8.1	9.4
	5% WB	MCDBR	20.1	20.7	22.3	23.8	21.4	15.8	11.0	10.7	12.0	13.0	16.2	20.4
MCWBR		9.4	9.0	8.5	9.3	8.6	6.0	4.6	4.6	5.2	5.8	7.2	9.4	
Clear-Sky Solar Irradiance	taub	0.452	0.449	0.495	0.535	0.600	0.637	0.561	0.546	0.547	0.524	0.517	0.483	
	taud	2.000	2.057	1.956	1.896	1.791	1.759	1.958	2.017	1.991	1.967	1.878	1.935	
	Ebn at Noon	252	264	258	249	232	222	239	244	240	239	229	235	
	Edh at Noon	51	51	59	64	70	72	59	56	56	55	57	52	
All-Sky Solar Radiation	RadAvg	1249	1535	1838	2031	1979	1613	1291	1336	1377	1445	1300	1168	
	RadStd	85	92	77	117	89	147	113	70	101	140	66	75	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page